

Jun-young Choi

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Research Agenda

- **Econometric Theory & Structural Estimation:** Selection and control-function methods under ill-conditioned designs; Regularization for stable two-stage corrections; Simulation-based likelihood for latent heterogeneity.
- **Structural Labor & Household Dynamics:** Wage formation with information frictions; Labor-market integration; Household behavior under policy and credit constraints.

Fields of Interest

Labor Economics, Econometrics

Education

Ph.D. Candidate in Economics , Korea University	Feb. 2026 (Expected)
<i>Advisor: Seik Kim</i>	
M.A. in Economics , Korea University	2019
B.A. in Economics (Minor: Mathematics), Kyungpook National University	2013

Research Papers

■ Accepted / Published

- [1] “Nowcasting South Korea’s 1990s Unemployment Rate with News Sentiment.” *Computational Economics*, (Accepted Dec. 2025).
(Method: Text-based sentiment measurement; High-dimensional time-series prediction.)
- [2] “Decomposing Unemployment with a Partially Labeled Mixture Model: Evidence from Censored Korean Data.” (in Korean) *The Korean Journal of Applied Statistics*, Forthcoming (Accepted Sep. 2025).
(Method: Partially labeled mixture modeling under censoring.)
- [3] “Developing a Short-Term Return Forecasting Model for the National Pension Fund Using KOSPI Market Indicators.” (in Korean, with Heontae Shin) *Journal of Budget and Policy*, Sep. 2025.
- [4] “Does Population Aging Exacerbate Income Inequality?” (in Korean, with Wonki Cho and Woojin Lee) *Journal of Korean Economic Development*, Dec. 2021.

■ *Under Review / Revision*

- [5] “Household Sorting and Welfare Portfolios in South Korea: A Nested Logit Approach.” *Minor Revision, Review of Economics of the Household*.
(Method: Multi-alternative program choice; Nested logit; Welfare portfolio decomposition.)

■ *Working Papers*

- [6] “Stabilizing Two-Stage Corrections for Multinomial Selection through Regularization.” 2025.
(Method: Regularized control function; Split-sample inference under ill-conditioned correction terms.)
- [7] “Potential Wages and the Value of a Domestic Credential: A Simulated Maximum Likelihood Approach for Foreign Workers in South Korea.” 2025.
(Method: Sample-selection corrected ordered probit; SML; Background-based instruments.)
- [8] “Gender Differences in Wages and Promotion Opportunities among Korean Managers.” (with Seik Kim) 2024.
(Data: KWMP. Method: Promotion and wage gaps controlling for job rank/position.)

Academic Research Experience

Research Associate, Development Futures Lab, Korea University 2025–2026

- Project: “A Reinforcement Learning Framework for Dynamic Risk Assessment in ODA Portfolios” (with Grace Choi).

Research Assistant, Prof. Jong-Wha Lee, Korea University Spring 2023

- Data mining and support for cross-country panel analysis on higher education quality.

Professional Research Experience

Senior Research Associate, Korea Culture and Tourism Institute 2021–2022

Researcher, Korea Institute of Health and Social Affairs 2020–2021

Researcher, Korea Institute of Public Finance 2019–2020

Work involved large-scale Korean administrative and survey microdata and policy evaluation reports.

Teaching Experience

Teaching Assistant, *Econometrics I* (Undergraduate), Prof. Sang Soo Park Spring 2018

Awards & Referee Service

Hakhyeon Award (Best Article of 2021), Korean Development Economics Association 2022
Referee Service: *Computational Economics* 2025

Academic Presentations

Special Session for Graduate Students, Joint Academic Conference on Economics Feb. 2019
Korean Association of Public Finance Annual Meeting Oct. 2018

Selected Policy Contributions

■ Reports (in Korean)

- [1] *A Study on Measures to Evaluate the Economic Effects of Culture, Sports, Tourism, and Content Industries*, KCTI, 2022.
- [2] *A Comprehensive Study of Happiness and Quality of Life in Korea*, KIHASA, 2021.

■ Public Service

- [1] Project person-in-charge for the Special Inspection of Government Business Subsidies (Head: Prof. Jin-Wook Choi), MOEF / KIPF, 2020.

Computational Skills & Languages

Core Languages & Tools: Python (JAX, TensorFlow), R, Stata, MATLAB, $\text{\LaTeX}$
Languages: English (Advanced), Korean (Native)